Weekly Report

Jyothsna Padma Kandarpa
National Institute of Technology Warangal

June 13, 2025

1 Introduction

The TIA converts the photodiode's output current into a voltage signal while maintaining low noise and high bandwidth performance.

The objective of this work is to design a TIA that produces a linear output voltage ranging from 0 to 3 V for an optical input of **0 dBm** (1 mW). The circuit is tailored to an InGaAs photodiode operating at 1550 nm, with careful selection of feedback and compensation components to meet performance goals.

2 Photodiode and Responsivity

An **Indium Gallium Arsenide (InGaAs)** photodiode is chosen for its superior response in the near-infrared region. Responsivity R, indicating how effectively light is converted to current, is defined as:

$$R = \eta \cdot \frac{q}{h\nu}, \quad R = 0.95 \text{ A W}^{-1}$$

For an optical power of 0 dBm (1 mW):

$$I_{PD} = R \cdot P = 0.95 \cdot 10^{-3} = 0.95 \text{ mA}$$

Parameter	Min.	Typ.	Max.	Note
Detection Range (nm)	1100		1650	
Power Supply, Vcc (V)	4.5		5.5	
Supply Current, Icc (mA)			35	No loads
Reverse Voltage, VR (V)		5		Absolute Maximum Ratings 30V
Differential Output Voltage, Vd (mV)		200	300	Pin=-30dBm, RL=100ΩΩ
Data Out Rise Time (10~90%) (ns)			4.5	1310nm
Data Out Fall Time (10~90%) (ns)			4.5	1310nm
Output Resistance (per side), Rout (Ohm)		50		
Bandwidth, BW (MHz)		155		1310nm
Saturation Power, Psat (dBm)	-3	0		1310nm
Sensitivity (dBm)			-37	1310nm, 155Mbps, BER10^-10
Dark Current (nA)		1	5	VR=5V, 25°C
Capacitance (pF)			0.75	VR=5V
Operating Temperature (°C)	-40		+85	

Figure 1: Specifications of InGaAs Photodiode used

3 TIA Circuit Design

Feedback Resistor

To achieve a 3 V output:

$$R_f = \frac{V_{\text{out}}}{I_{PD}} = \frac{3}{0.95 \times 10^{-3}} \approx 3.16 \text{ k}\Omega$$

A standard 3.3 kΩ resistor is used in practice.

Capacitance Considerations

The total input capacitance C_T seen by the op-amp includes:

- Photodiode capacitance: $C_{PD} = 1$ pF
- Op-amp input capacitance: $C_{in} = 3$ pF
- Parasitic capacitance: $C_P = 1$ pF

$$C_T = 1 + 3 + 1 = 5 \text{ pF}$$

This value influences the bandwidth and stability of the TIA.

Simulation Attempt

An initial simulation attempt was made using LTSpice. The op-amp was modeled with default parameters, and the photodiode was represented as a current source with parallel capacitance. I made mistake in the way i placed the current source which was not giving me correct output. on discussion with ashutosh, he came to know i have not designed amplifier circuits before so he told me work on the OOK set up as a good practice.

4 Next Week's Task: OOK Free-Space Optics

To enhance circuit design skills, the next task is to build an **OOK (On-Off Keying)** free-space optical communication setup. This involves modulating a laser with a binary signal, transmitting it through open space, and detecting the signal with a photodiode-TIA receiver.